LUCAS MOUGEOT

Systems Engineer | Full-Stack & AI-Driven Platforms

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SUMMARY

Full-stack systems engineer who designs and ships reliable, user-facing platforms that transform complex workflows into intuitive, production-grade software. Experienced building TypeScript/Next.js applications backed by PostgreSQL and containerized services. Comfortable owning features end-to-end and partnering with product and design to deliver scalable, high-quality systems in fast-moving environments.

EXPERIENCE

Integration Engineer — EZ Systems (Full-Time) Feb 2025 – Aug 2025

- Led systems integration efforts across AI, robotics, and sensor-driven platforms in regulated deployment environments.
- Designed unified QA and deployment pipelines integrating AI inference, calibration workflows, and CI/CD automation to ensure reproducible, production-ready system validation.
- Developed geometric processing and motion-planning software translating CAD models into executable machine instructions for autonomous systems.
- Authored internal AI integration standards and contributed to federal grant proposals on explainable AI systems.

Founding Systems Engineer — Stealth Compliance Platform 2024 – Present

- Owned end-to-end delivery of full-stack features across React/Next.js frontend, API services, background workers, and PostgreSQL data models.
- Built a workflow automation platform translating complex regulatory and operational logic into intuitive, structured user experiences that reduced manual review and improved decision consistency.
- Developed schema-driven form and workflow engines enabling dynamic validation, automated routing, and AI-assisted recommendations.
- Integrated LLM-powered retrieval and hybrid search (vector + full-text) to enhance automation, surface contextual guidance, and reduce user friction.
- Established comprehensive testing practices including concurrency, fuzz, and end-to-end validation to ensure production reliability.
- Designed scalable relational and graph-based data models supporting versioned workflows, high-performance query patterns, and reliable production growth.
- Deployed and maintained multi-service Docker architecture (PostgreSQL, Redis, BullMQ) with CI/CD and automated regression testing.

Graduate Researcher — Materials Science (NASA-Funded Capillary Devices) 2023 – 2024

- Designed and fabricated porous ceramic devices enabling passive capillary transport in microgravity environments for space agriculture.
- Developed 36 ceramic formulations optimizing capillary rise, stability, and manufacturability.
- Built MATLAB computer-vision pipelines for pore segmentation, morphology quantification, and statistical ranking of candidate formulations.
- Characterized microstructure and composition using SEM/EDS; documented processing–structure–property relationships.

Undergraduate Researcher — Computational Geometry (NSF-Funded) 2020 – 2022

- Implemented bounded-degree planar graph constructions in modern C++ using CGAL under strict geometric constraints.
- Designed locality-preserving spatial algorithms for dense and heterogeneous point datasets.
- Contributed to peer-reviewed ACM Journal of Experimental Algorithmics publication.

SELECTED PROJECTS

Automated Lithium-Ion Battery Pack Assembler Senior Design Capstone

- Designed and implemented an automated lithium-ion battery pack assembly system using an AutomationDirect Do-More PLC.
- Developed ladder logic coordinating motion sequencing, interlocks, fault handling, and safe state transitions.
- Integrated an RP Series robotic dispenser via RS-232 serial communication to initiate external job file execution.
- Implemented PWM-based servo motor control for precise positioning and timing.
- Delivered a deterministic, repeatable automation pipeline with full I/O mapping and bring-up procedures prior to generative AI tools.

EDUCATION

University of North Florida

M.S. Materials Science & Engineering (Thesis-Based) — GPA: 4.00 / 4.00

- Thesis: Development and Characterization of Ceramic Devices for Capillary Action in Microgravity for the Purpose of Growing Vegetables in Space (NASA-Funded)

B.S. Electrical Engineering — GPA: 3.93 / 4.00

B.A. Interdisciplinary Studies: Morphology of Futurism — GPA: 4.00 / 4.00

Concentrations: Mechanical Engineering, Materials Science, Chemistry, Physics, Computing, Mathematics, Mandarin Chinese

PUBLICATIONS & PRESENTATIONS

- Bounded-Degree Plane Geometric Spanners in Practice — ACM Journal of Experimental Algorithmics (Peer-Reviewed)
- Highly Sensitive and Stable pH Sensor Electrodes of TiN Fabricated Using HiPIMS with Kick — AVS 69 Poster, 2023
- Interactive Tool for Bounded-Degree Plane Geometric Spanners — SoCG Media Exposition, 2021

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C++
Frontend: React, Next.js, TailwindCSS
Backend: Node.js, API design, background workers
Infrastructure: Docker, PostgreSQL, Redis, CI/CD pipelines
Data & AI: LLM integration, hybrid search (vector + full-text), distributed systems
Tools: GitHub Actions, Supabase, BullMQ, Playwright